

Level 1 — 25 marks/question**L1-Q1. (E) Leap**

j

CODE

```
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        int year;
        Scanner sc = new Scanner(System.in);
        year=sc.nextInt();
        if(year%4==0){
            System.out.println("It is a leap year");
        }
        else{
            System.out.println("It is not a leap year");
        }
    }
}
```

INPUT

2024

OUTPUT

It is a leap year

L1-Q2. (E) Fibonacci

j

CODE

```
import java.util.Scanner;
import java.util.Arrays;
class Main{
    public static void main(String[] args){
        int n,sum=0;
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a number");
        n=sc.nextInt();
        int first=0,next=1;
        int arr[] = new int[10];
        for (int i = 0;i<=n;i++){
            arr[i]=first;
            first=next;
            next=arr[i]+next;
        }
        for(int i=0;i<=n;i++){
            if(i%2==0)
                sum+=arr[i];
        }
        System.out.println("The sum is "+sum);
    }
}
```

INPUT

8

OUTPUTEnter a number
The sum is 33

Report

L1-Q4. (E) Frequency

j

CODE

```
import java.util.Scanner;
class main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n;
        int visited=-1;
        System.out.println("Enter the no of elements");
        n=sc.nextInt();
        int arr[] = new int[n];
        for(int i = 0; i<arr.length;i++){
            arr[i]=sc.nextInt();
        }
        System.out.println("Element | Frequency");
        int[] visitedArr = new int[arr.length];
        for(int i=0;i<arr.length;i++){
            if(visitedArr[i]==visited){
                continue;
            }
            int count = 1;
            for(int j=i+1;j<arr.length;j++){
                if(arr[i]==arr[j]){
                    count++;
                    visitedArr[j]=visited;}
            }
            visitedArr[i]=count;
        }
        for(int i=0;i<arr.length;i++){
            if(visitedArr[i]!=visited){
                System.out.println(arr[i] + "occurs " + visitedArr[i]);
            }
        }
    }
}
```

INPUT

4 1 1 2 2

OUTPUT

```
Enter the no of elements
Element | Frequency
1occurs 2
2occurs 2
```

L1-Q3. (E) Sort Strings**j****CODE**

```
import java.util.Scanner;
import java.util.Arrays;
class Main{
    public static void main(String[] args){
        int n;
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the no of elements");
        n=sc.nextInt();
        String arr[]= new String[n];
        System.out.println("Enter the strings ");
        for(int i=0;i<n;i++){
            arr[i]=sc.next();
        }
        Arrays.sort(arr);
        char ch;
        System.out.println("Enter your choice A or D");
        ch=sc.next().charAt(0);
        if(ch == 'A'){
            for(int i=0;i<n;i++){
                System.out.println(arr[i]);
            }
        }
        else if(ch == 'D'){
            for (int i=n-1;i>=0;--i){
                System.out.println(arr[i]);
            }
        }
        else{
            System.out.println("Invalid Choice");
        }
    }
}
```

INPUT

```
4
Banana Carrot Apple Jack
A
```

OUTPUT

```
Enter the no of elements
Enter the strings
Enter your choice A or D
Apple
Banana
Carrot
Jack
```